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Creating Value, Leveraging Knowledge
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Department of

Electronics and Communication Engineering

Mini Project Report
on
Noise Cancellation in ECG Signal

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the project entitled **"NOISE CANCELLATION IN ECG SIGNAL"** is a bonafide work carried out by the student team of **"Aksheta B Patel (02FE22BEC005), Bhagyashree R Jainapur(02FE22BEC016)**, **Chandrashekhar Bhavi (02FE22BEC020), Pruthviraj U Banoshi (02FE23BEC058)**. The project report has been approved as it satisfies the requirements with respect to the mini project work prescribed by the university curriculum for B.E. (V Semester) in the Department of Electronics and Communication Engineering of KLE Technological University, Dr. M. S. Sheshgiri CET Belagavi campus for the academic year 2024-2025.

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- The Project Team

ABSTRACT

Electrocardiography (ECG) is an essential tool for diagnosing and monitoring heart health. However, ECG signals are often affected by various types of noise, including power line interference, baseline wander, muscle artifacts, and electrode motion artifacts. These disturbances can mask key features in the signal, such as P-waves, QRS complexes, and T-waves, which are crucial for accurate diagnosis. To maintain the integrity of ECG signal processing and analysis, effective noise cancellation techniques are vital. This paper examines advanced methods for noise removal in ECG signals, including adaptive filtering, wavelet transform, and machine learning-based approaches. Adaptive filters like Least Mean Squares (LMS) and Recursive Least Squares (RLS) are commonly used for real-time noise suppression. The wavelet transform offers multiresolution analysis, making it especially useful for separating non-stationary noise components. In recent years, machine learning and deep learning models have shown significant promise in denoising ECG signals, leveraging large datasets to improve accuracy.

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. List of Abbreviations

Abbreviations	Description
WBS	Work Breakdown Structure
SNR	Signal to Noise Ratio
HRV	Heart Rate Variability
LMS	Least Mean Square
NLMS	Normalized least Mean Square
RLS	Recursive least squares
PSD	Power spectrum density
PLI	Power line interference

Chapter 1

1 Introduction

Electrocardiograms (ECGs) are used to record the heart's electrical activity and are fundamental tools in diagnosing cardiovascular conditions. Accurate detection and measurement of features within ECG signals are vital for effective clinical evaluation. However, the presence of power line interference (PLI), typically at 50 Hz, often contaminates ECG recordings and diminishes signal quality if left untreated. Removing this interference is a critical step in ensuring precise signal interpretation and has become a significant focus of research in signal processing.

A variety of methods have been explored to address the issue of PLI in ECG signals. Among these, notch filters based on Fourier analysis are widely applied, including designs using finite or infinite impulse response. For example, multinotch filtering approaches have been utilized to mitigate harmonic interference, while Spectro-temporal filtering methods have shown improvements in the signal-to-noise ratio (SNR). Adaptive notch filtering techniques, designed to specifically target and suppress harmonic components, have also been proposed.

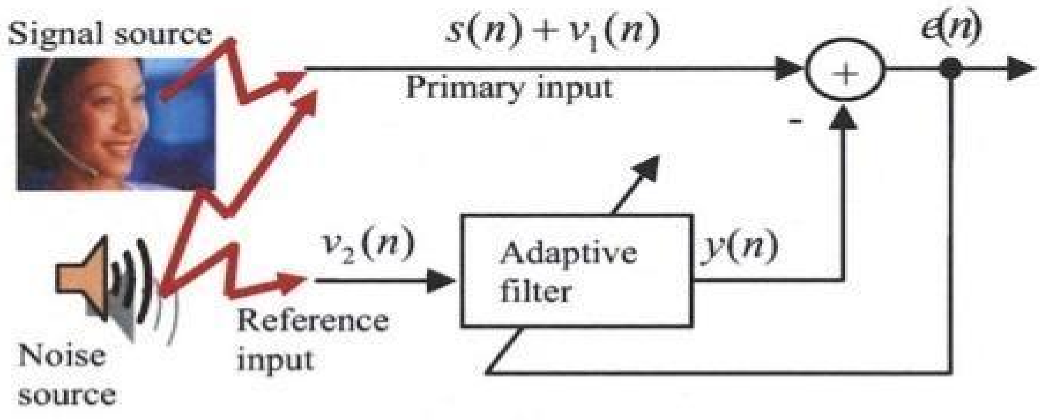

Figure 1.1 Primary reference

Several additional techniques have been developed to improve the accuracy of ECG signal analysis. Adaptive filtering methods based on least mean square algorithms, as well as approaches employing adaptive Fourier transforms, Kalman filters, nonlinear Bayesian filtering, and eigenvalue decomposition, have demonstrated their utility in ECG noise suppression. Recently, artificial intelligence (AI) techniques have emerged as powerful tools in this domain. For example, adversarial networks have been employed to achieve significant improvements in SNR for denoised signals, while deep learning frameworks have been developed to enhance ECG classification accuracy.

1.1 Motivation:

Noise cancellation in ECG signals is essential because these low amplitude signals are highly susceptible to various noise sources, such as powerline interference, muscle activity, motion artifacts, and baseline wander, which can distort the and hind signal er accurate diagnosis. Traditional fixed filters often fail to handle the non-stationary nature of noise or risk distorting clinically significant information. Adaptive filters, such as LMS and RLS, dynamically adjust their coefficients in real time, effectively suppressing noise while preserving the integrity of the ECG signal. This capability makes adaptive filtering ideal for applications requiring clean ECG signals, such as real-time monitoring, wearable devices, and automated diagnostics.

1.2 Objectives:

- Minimize noise and artifacts in ECG signals.
- Preserve the integrity of the ECG waveform.
- Adapt dynamically to non-stationary noise.
- Enable real-time noise cancellation for monitoring devices.
- Enhance diagnostic accuracy in cardiac analysis.
- Handle various noise sources like powerline interference and motion artifacts.

1.3 Literature Survey:

Electrocardiograms (ECGs) are used to record the heart's electrical activity and are fundamental tools in diagnosing cardiovascular conditions. Accurate detection and measurement of features within ECG signals are vital for effective clinical evaluation. However, the presence of power line interference (PLI), typically at 50 Hz, often contaminates ECG recordings and diminishes signal quality if left untreated. Removing this interference is a critical step in ensuring precise signal interpretation and has become a significant focus of research in signal processing.

A variety of methods have been explored to address the issue of PLI in ECG signals. Among these, notch filters based on Fourier analysis are widely applied, including designs using finite or infinite impulse response. For example, multinotch filtering approaches have been utilized to mitigate harmonic interference, while Spectro-temporal filtering methods have shown improvements in the signal-to-noise ratio (SNR). Adaptive notch filtering techniques, designed to specifically target and suppress harmonic components, have also been proposed.

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1.4 Problem statement:

Develop a real-time noise cancellation system for ECG signals to enhance heart rate monitoring. The system will analyse the ECG data, removing unwanted noise and interference, ensuring accurate and reliable heart activity readings. This will improve diagnostic accuracy, help detect potential heart conditions early, and reduce the risk of misinterpretation or false diagnoses.

Chapter 2

Project planning

Project planning is critical for ensuring project success by clearly defining objectives, identifying tasks, and setting timelines. It enables efficient resource allocation, risk management, and helps to maintain focus and alignment among team members. Additionally, it enhances communication, allowing for smooth coordination, and ensures that milestones are met within budget and scope.

2.1 Gantt chart:

Figure 2.1 Gantt chart

A Gantt chart is a visual project management tool that represents the timeline of a project. It displays tasks or activities as horizontal bars along a timeline, with the length of each bar indicating the duration of a task. The Gantt chart helps project managers track progress, manage deadlines, and visualize how tasks overlap or depend on one another. It is widely used for scheduling, resource allocation, and ensuring timely completion of projects.

2.2 Work Breakdown Structure (WBS):

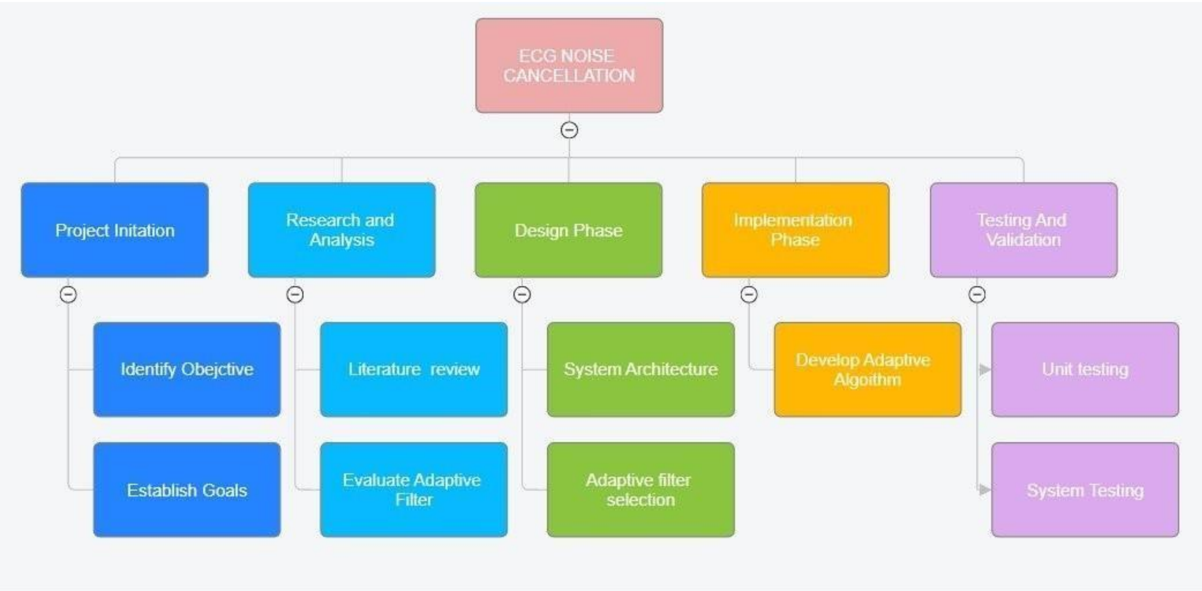

Figure 2.2 Work Breakdown Structure

A Work Breakdown Structure (WBS) is a project management tool that divides a project into smaller, manageable components or tasks. It organizes the work into a hierarchy, making it easier to plan, assign resources, and track progress. The WBS helps clarify the scope of the project, ensuring that all required work is identified. It also enhances communication among team members and stakeholders, assigns responsibilities, and improves time and resource management. By breaking down complex projects into manageable parts, it makes it easier to estimate costs, schedule timelines, and monitor progress.

Chapter 3

Design specifications

3.1 Input specifications:

- Signal Source: Raw ECG data having noise
- Noise Types:

  Powerline interference (50/60 Hz).

  Baseline wander (low-frequency drift due to movement or respiration).

  EMG noise (high-frequency artifacts from muscle activity).

  Motion artifacts (signal distortion due to electrode displacement).

- Sampling Rate: Typically, 250 Hz, 500 Hz, or 1 kHz, depending on the system. • Signal Range: Amplitude typically in millivolts (mV).

3.2 Output specifications:

- Cleaned ECG Signal:

  Retains critical diagnostic features: P-wave, QRS complex, and T-wave.

  Free from significant noise artifacts.

- Improved Metrics:

  Higher Signal-to-Noise Ratio (SNR).

  Preservation of signal morphology for diagnostic accuracy.

- Suitability for Analysis:

  Usable for automated heart rate variability (HRV) analysis.

  Reliable for arrhythmia and other cardiac condition diagnostics

3.3 Functional block diagrams:

An adaptive filter may be understood as a self-adjusting filter that changes its coefficients to minimize the error. The most crucial property of adaptive filter is that it can work for known framework as well as unknown framework. The proposed block diagram of the adaptive noise cancellation for ECG signal.

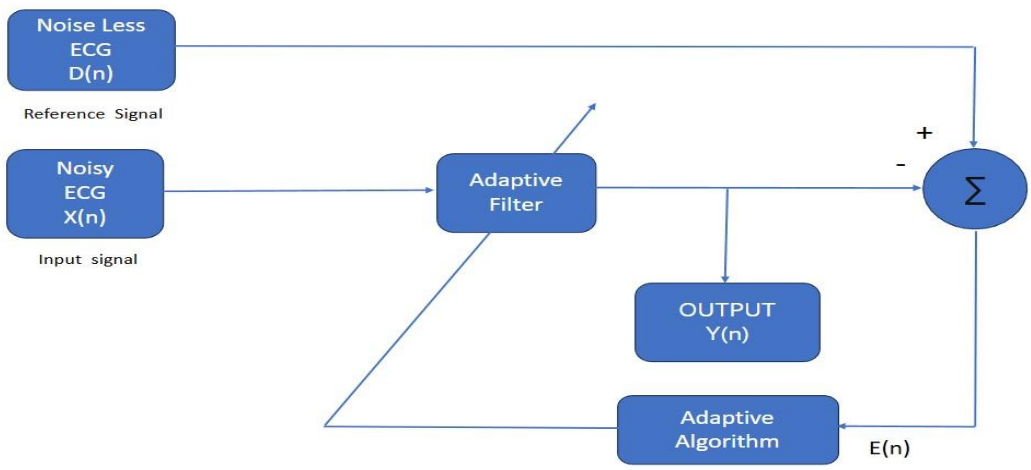

Figure 3.3 Functional block diagrams

It is considered as the desired signal as D(n) which is noise less ECG signal. For input X(n) . we have chosen an impulsive noise affected ECG signal. The error signal which is calculated by taking the difference between the desired signal and the output signal collected by passing the input signal through an adaptive filter is denoted by E(n).

$$E(n)= D(n)-Y(n)$$

The adaptive channel changes its weights as indicated by the error signal. At whatever point the error will be least, optimal point will be reached and at this case of time we will get our clean signal without impulsive noise. The output signal is given by Y(n).

$$Y(n)=W(n)X(n)^T$$

Chapter 4

Results and discussions

4.1 Clean ECG Signal using LMS, NLMS, RLS Output signal:

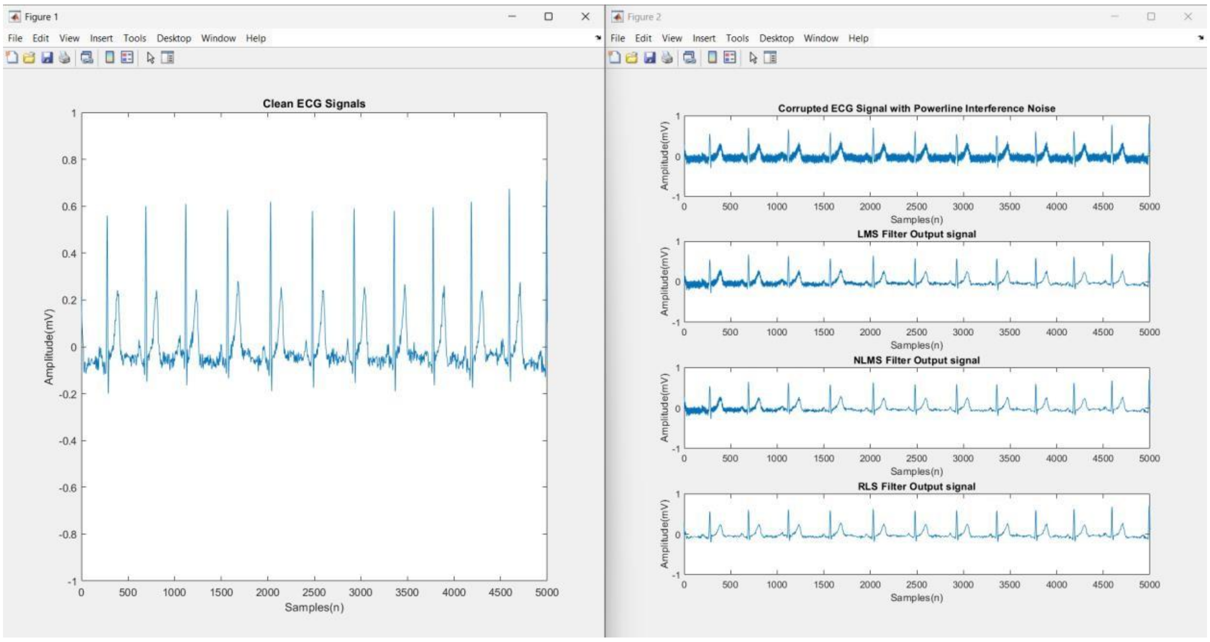

Figure 4.1Clean ECG Signal using LMS, NLMS, RLS Output signal

1. Left Image: Clean ECG Signal

   - Displays the ideal ECG waveform without noise, showing distinct R-peaks.

2. Right Image: Noisy and Filtered Signals

   - Top Plot: ECG signal corrupted with powerline interference, making analysis difficult.

   - LMS Filter: Reduces some noise; R-peaks are clearer but residual noise remains.

   - NLMS Filter: Improves noise suppression over LMS, with clearer ECG features.

- RLS Filter: Provides the best noise removal, with a signal closest to the clean ECG.

3. Conclusion:

- Adaptive filters effectively reduce powerline interference.
- The RLS filter shows superior performance in noise cancellation and signal clarity.

4.2 Output signal of power spectrum density:

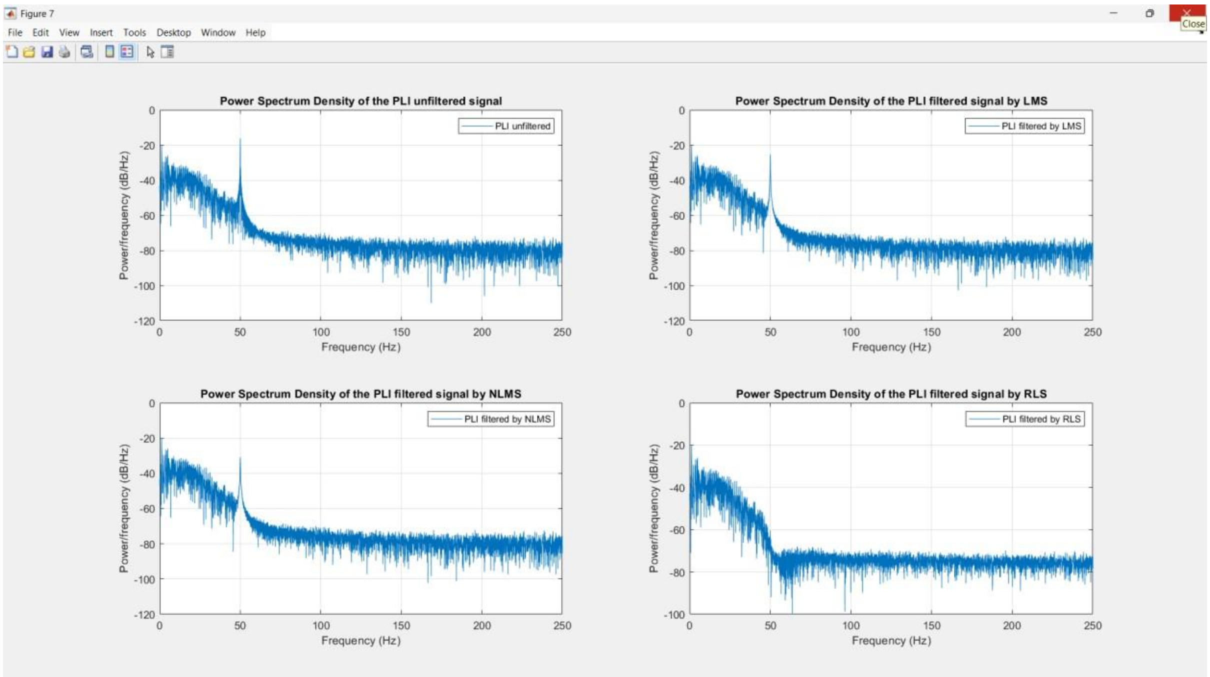

Figure 4.2 Output signal of power spectrum density

This image presents four plots showing the power spectrum density (PSD) of a signal affected by power line interference (PLI) and its filtering using different algorithms.

1. Top-Left Plot: Displays the PSD of the unfiltered PLI signal. A strong peak is visible around 50 Hz (likely representing the interference frequency) with high power.

2. Top-Right Plot: Shows the PSD of the signal after filtering using the Least Mean Squares (LMS) algorithm. The 50 Hz peak is significantly reduced, indicating effective filtering.
3. Bottom-Left Plot: Illustrates the PSD of the signal filtered using the Normalized LMS (NLMS) algorithm. Similar to LMS, the 50 Hz component is mitigated, with slight differences in the residual noise.
4. Bottom-Right Plot: Depicts the PSD of the signal filtered using the Recursive Least Squares (RLS) algorithm. The interference at 50 Hz is substantially reduced, often more effectively than LMS and NLMS.

Overall, these plots compare the performance of LMS, NLMS, and RLS algorithms in suppressing PLI in terms of PSD.

4.3 Output of difference between adapter filter and notch filter:

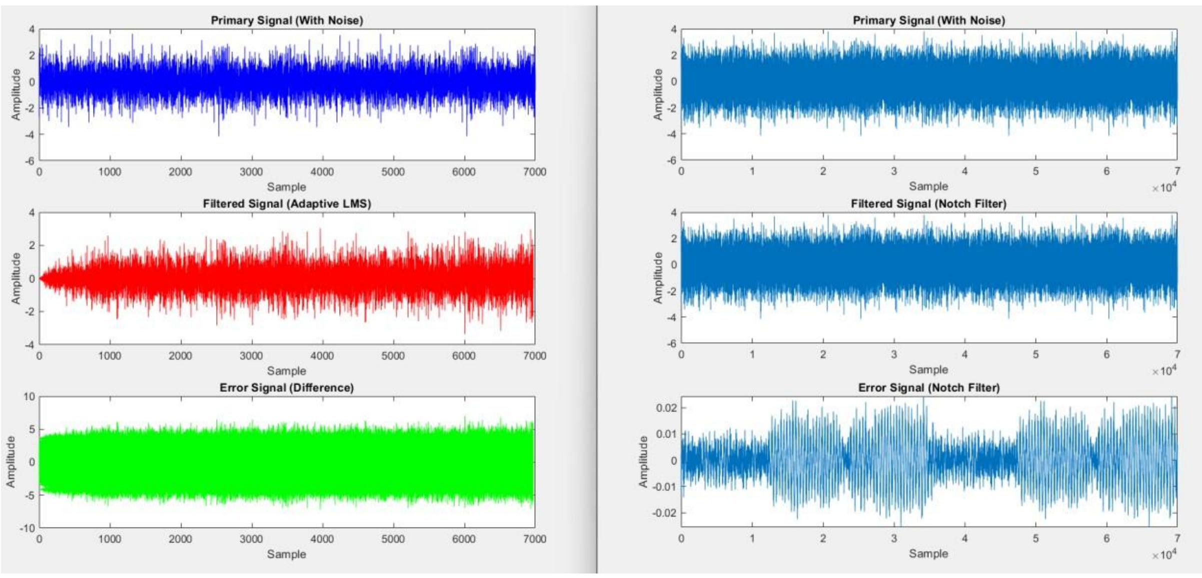

Figure 4.3 Output of difference between adapter filter and notch filter

1. Enhanced Diagnostic Accuracy in Healthcare
   Adaptive notch filters improve the clarity of ECG signals by effectively removing noise, leading to more accurate cardiac diagnosis and better clinical decisionmaking.

2. Integration into Wearable and IoT Devices
   Compact and energy-efficient adaptive filters are ideal for wearables like smartwatches and portable ECG devices, enabling real-time monitoring in dynamic environments.

3. Personalized and Adaptive Filtering
   Filters can be designed to adjust dynamically to each patient's unique physiological signals and specific noise patterns, improving overall reliability and performance.

4. Hybrid with AI and Machine Learning
   Integration with AI enables noise source identification, learning from patient specific data, and adaptive behaviour for complex noise environments, enhancing functionality.

5. Applications in Remote Monitoring and Telemedicine
   Noise-filtered ECG data can be transmitted in real time through IoT systems, improving the quality of remote cardiac monitoring and emergency care diagnostics.

Conclusion

Noise cancellation in ECG signals is a crucial step to ensure accurate analysis and diagnosis of heart-related conditions. ECG signals often encounter various types of noise, including powerline interference, baseline wander, and motion artifacts, which can distort the signal and obscure critical features. To address this, advanced signal processing techniques such as adaptive filtering, wavelet transform, and empirical mode decomposition are commonly employed. These methods aim to isolate and remove noise while preserving the integrity of the original ECG waveform. The application of digital filters, particularly low-pass and high-pass filters, is another effective strategy to suppress unwanted frequencies and enhance the signal quality.

The effective noise cancellation in ECG signals significantly improves the reliability of clinical diagnoses and automated heart condition detection systems. By removing noise and preserving essential signal characteristics, these techniques enhance the clarity of PQRST complexes and other diagnostic markers. Advanced machine learning and deep learning models further augment the noise cancellation process by learning complex noise patterns and adapting to various signal conditions. Overall, noise reduction not only ensures better diagnostic accuracy but also paves the way for efficient remote monitoring systems and wearable health devices.

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